

Advanced (Habanero)

- Q1.** Which of the following is an equivalent ratio to 3 : 5?
(A) 6 : 10
(B) 9 : 12
(C) 12 : 15
(D) 9 : 15
(E) 15 : 20
- Q2.** Solve for x in the proportion $\frac{x}{4} = \frac{9}{12}$
(A) 3
(B) 6
(C) 9
(D) 12
(E) 15
- Q3.** A recipe for making cookies calls for a ratio of 2 : 3 for flour to sugar. If you need 6 cups of flour, how many cups of sugar do you need?
(A) 6 cups
(B) 7 cups
(C) 8 cups
(D) 9 cups
(E) 10 cups
- Q4.** A group of friends want to split some money in a ratio of 2 : 3 : 5. If they have a total of \$200, how much will the person with the ratio 5 get?
(A) \$40
(B) \$50
(C) \$60
(D) \$80
(E) \$100
- Q5.** A car travels 250 miles in 5 hours. How many miles will it travel in 8 hours at the same rate?
(A) 300 miles
(B) 350 miles
(C) 400 miles
(D) 400 miles
(E) 450 miles
- Q6.** If $\frac{a}{b} = \frac{2}{3}$ and $a = 8$, what is the value of b ?
(A) 10
(B) 12
(C) 15
(D) 18
(E) 20
- Q7.** A bicycle is on sale for 15% off its original price of \$80. How much will you pay for the bicycle?
(A) \$60
(B) \$65
(C) \$68
(D) \$70
(E) \$75
- Q8.** A bakery is having a sale on bread. A loaf of bread normally costs \$2.50, but it is on sale for 10% off. How much will you pay for the loaf of bread?
(A) \$2.25
(B) \$2.30
(C) \$2.40
(D) \$2.45
(E) \$2.50

Open Ended Questions (FRQ)

- Q9.** Tom has been saving money for a new bike and has \$120 in his savings account. He wants to buy a bike that costs \$180. His parents have agreed to give him an allowance of \$5 per week for doing chores. How many weeks will it take Tom to have enough money to buy the bike? Show your work and explain your reasoning.
- Q10.** A water tank can hold 1200 gallons of water. Due to a leak, the tank is losing water at a rate of 2% per hour. How many gallons of water will the tank have after 5 hours? Show your work and explain your reasoning.

Chef's Tips

- Remind students to check their work and explain their reasoning
- Encourage students to use proportions to model real-world situations
- Consider providing extra support for students who struggle with ratios and proportions